

Exploring cognitive models to augment explainability in Deep Knowledge Tracing

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Adaptive learning systems allow a personalized adaptation based on the characteristics of the student. Tracing the progress of knowledge and skills during the learning process through cognitive models is essential so that these systems can make appropriate decisions when carrying out personalization. This is the objective of Knowledge Tracing, which studies how to infer a cognitive model from the answers given to a sequence of questions or exercises. The incorporation of Deep Learning techniques in this field has given rise to Deep Knowledge Tracing (DKT) which usually has excellent predictive outcomes. The problem is that this increase in accuracy comes with a lack of explainability since Deep Learning models can be considered black boxes from which it is difficult to build interpretations or explanations. By contrast, traditional Knowledge Tracing methods are based on underlying learning models and provide a solid basis for explainability. In this paper we describe an ongoing research to build DKT models with a good trade-off between accuracy and explainability. To this end, we propose to use a loss function based on a mixup approach where the ground truth is a mix between the dataset labels and the predictions of a surrogate explainable model. The approach has potential to improve, not only explainability through the use of the surrogate, but also accuracy thanks to regularization effects. We will validate the approach by exploring, for different cognitive models, the trade-off curve that is obtained by plotting accuracy against explainability for different mixup values.

Additional Key Words and Phrases: personalized learning systems, scrutable user models, explainability, deep knowledge tracing (DKT), cognitive models

ACM Reference Format:

Concha Labra and Olga C. Santos. 2023. Exploring cognitive models to augment explainability in Deep Knowledge Tracing. In *Adjunct Proceedings of the 31st ACM Conference on User Modeling, Adaptation and Personalization (UMAP '23 Adjunct)*, June 26–29, 2023, Limassol, Cyprus. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3563359.3597384>

1 INTRODUCTION

The objective of Knowledge Tracing is to track the progress of a student in terms of the knowledge or skills developed throughout a learning process. In [17] the problem of Knowledge Tracing was posed as a task that could be approached using Recurrent Neural Networks (RNNs) giving rise to the field of Deep Knowledge Tracing (DKT).

The different DKT models that have been proposed since then have shown excellent performance in terms of accuracy [20] although they suffer from the typical lack of explainability associated with Deep Learning methods. Many alternatives have been proposed to improve Deep Learning explainability, being surrogates a special interesting option because they are easy to understand and implement. A surrogate is a model that is trained with the output of the black-box model in order to be able to emulate its behavior [12]. The surrogates have to be explainable themselves because their purpose is, not to provide insights about the data, but explanations about the black box. Traditional Knowledge Tracing models are good surrogate candidates due to its intrinsic explainability. Once they are fit, the

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Manuscript submitted to ACM

recovered parameters can be used to conform a cognitive model with relevant characteristics such as knowledge and skills that are able to support interpretation and explanations.

One example of this is the Additive Factor Model (AFM) [3] [16], a logistic regression model that defines parameters related to the prior knowledge of students, as well as the difficulty and learning rate for each skill. AFM is based on the idea that the probability of answering an item correctly is proportional to a combination of the ability of the student, the difficulty of skills involved in the item and the amount of learning gained from each attempt.

Giving up this explainability when using DKT models is a concern in a field such as adaptive learning where it is especially important to be able to offer feedback, explain the decisions taken to personalize adaptation and, ideally, allow users to correct reasoning and system assumptions where needed as in the case of scrutable user models [18].

The goal of this ongoing research is to explore the possibility of augmenting explainability in DKT models without overlooking accuracy. To this end we propose to use a loss function based on a variant of the *mixup* method described in [26]. In our case, we use a mix between the ground truth y given by the dataset and the predictions of an explainable model $\hat{y}$, with a hyper-parameter λ that controls the proportion:

$$y_{mixup} = \lambda * y + (1 - \lambda) * \hat{y}$$

Similar approaches like *noisy labels*, where the learner instead of being trained with the true labels uses labels that have been modified with some small probability, have shown high robustness [14]. If our hypothesis is correct, the mixup approach might allow to obtain DKT models that are explainable through surrogates presenting, at the same time, good accuracy performance for some λ values. To validate the approach we will explore curves that show explainability against accuracy for different λ values looking for pairs with a good trade-off.

The structure of the paper is as follows: section 2 provides a background on traditional and DKT variants of Knowledge Tracing, section 3 presents the proposed approach and methodology based on the trade-off curves obtained using a mixup loss function, section 4 explains the experimental setup to validate the approach and, finally, section 5 describes ongoing work and research to be completed in order to get results through a working implementation

2 BACKGROUND

2.1 Traditional Knowledge Tracing and explainability

Traditional KT, based mainly on Bayesian Networks and Factor Analysis models, are explainable because they draw on fields as diverse as education, cognitive science and psychometrics in order to model the cognitive state of the students.

Some of the most cited Factor Analysis models in the literature are Item Response Theory (IRT), Additive Factor Models (AFM), Performance Factor Analysis (PFA) and Knowledge Tracing Machines (KTM) [2]. All of them are based on underlying learning principles in such a way that, once the model is fit, recovered parameters can be used as a basis for a cognitive model that is able to explain the evolution of the student in terms of knowledge and skills. In IRT, for instance, the relevant parameters are student ability and item difficulty [6] while, in the case of AFM, the factors considered are the ability of the student, the difficulty of skills involved in the item and the amount of learning gained from each attempt [3].

2.2 DKT

There are recent reviews on KT that have included a thorough overview of DKT [2] alongside other reviews specialized in DKT that provide a detailed analysis and categorisation of the main models and approaches [20].

The large number of variants and improvements that appeared after the first DKT model [17] can be classified in 4 categories according to whether they are based on Recurrent Neural Networks (RNN), memory networks, attention mechanisms or graph structures [20].

[9] introduces a comprehensive python based benchmark platform, pyKT, allowing for thorough evaluations and comparison across DKT methods and applying consistent data preprocessing on well known datasets. The benchmark implements many DKT models including the original RNN approach presented by [17] that was followed for more sophisticated approaches such as DKT+ that makes use of regularization terms to enhance the predictions consistency [25]. It is worth noting the AKT model that in pyKT benchmarks outperforms the rest in many scenarios thanks to the use of attention mechanisms [7].

Other variants are included as well in pyKT such as DKVMN that uses dynamic key-value memory networks to exploit the relationships between latent KCs [27], GKT that uses graphs to model the relations between knowledge concepts and reformulates the KT task as a time-series node-level classification problem [13], HawkesKT that introduces Hawkes process to model temporal cross effects [23], IEKT that uses estimations of individual students' cognition [10] or KQN that uses neural networks to encode student learning activities into knowledge state and skill vectors [8].

Additionally, we can mention other variants that are also included in pyKT such as LPKT that models students' learning process to monitor their knowledge state [19], SAKT that uses attention mechanisms [15], SAINT that uses attention and transformers [5], SKVMN that makes use of memory networks [1], qDKT that uses the question ID as the input [21] and, finally, Deep-IRT that follows an item response theory approach [24].

3 APPROACH AND METHODOLOGY

Most current DKT models have been designed to perform well in terms of accuracy but not in terms of explainability. We look instead for an approach that takes into account both criteria trying to find a suitable trade-off between them. To this end, our proposal is based on the use of a loss function that uses a mixup between a term that looks after minimization of the error and another that looks after producing the same predictions than the explainable model. We plan to plot a trade-off curve displaying accuracy against explainability for different values of a hyperparameter that controls the mixup and explore the behavior of this curve in order to answer the next research questions:

- RQ1: Does an optimal trade-off exist?
- RQ2: How the behavior of the curve and the optimal trade-offs (if any) differ for different models?

The proposed methodology is based on training the DKT model following these steps:

- (1) Choose a value for λ
- (2) Get the predictions of the DKT model using the mixup loss function with this λ
- (3) Measure the accuracy for these predictions
- (4) Train the surrogate model with these predictions
- (5) Measure the explainability, i.e. how well the surrogate replicates the predictions of DKT
- (6) Update the trade-off curve with these accuracy and explainability values
- (7) Go to step 1 until until a certain condition is satisfied

4 EXPERIMENTAL SETTING

4.1 Algorithms

In this ongoing research we use as baselines the pyKT implementations of the original DKT algorithm, in addition to DKT+ and AKT. DKT+ is a variant that introduces regularization terms to enhance the consistency in predictions whereas AKT uses a monotonic attention mechanism to capture short-term dependencies on the past. We have chosen these models taking into account pyKT benchmark results where AKT outperforms the rest of DKT models and DKT+ and vanilla DKT beat most of them [9].

Factorization machines (FMs), a model that encompasses several other as special cases, is used as a base to explore Additive Factor Models (AFMs) [3], Performance Factor Analysis (PFA) [16] and variants of Item Response Theory (IRT) [11]. To implement FMs, we use the *KTM framework* to recover different learning models by selecting the appropriate features [22] [4] and use them to obtain cognitive profiles like the one shown in figure 1.

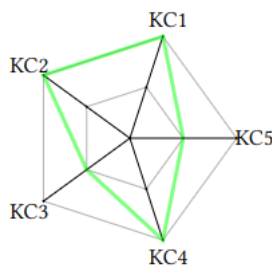


Fig. 1. Cognitive profile of a user mastering KC_1 , KC_2 and KC_4 knowledge components.

4.2 Datasets

We are working with these datasets included among the ones that can be preprocessed with pyKT:

- Algebra2005 is a dataset from the KDD Cup 2010 EDM Challenge. It contains 13-14 year old students' responses to Algebra questions with detailed step-level responses. Includes 809,694 interactions, 574 students, 210,710 questions and 112 KCs.
- ASSISTments2009, ASSISTments2012, ASSISTments2015 and ASSISTments2017 are datasets with information about math exercises collected from the online tutoring ASSISTments. They have been used as standards in benchmarks for KT methods over the last years.
- Bridge2006 was used in the KDD Cup 2010 EDM Challenge. It contains 3,679,199 interactions, 1,146 students, 207, 856 questions and 493 KCs.
- Ednet is a large-scale hierarchical student activity data set collected by the Santa platform. It includes 131,317,236 interactions of 78,4309 students.
- NIPS34 was used in the NeurIPS 2020 Education Challenge. It contains students' answers to multiple-choice diagnostic math questions collected from the Eedi platform. It contains 1,382,727 interactions, 948 questions, and 57 KCs.
- POJ has programming exercises collected from the Peking coding practice platform. It contains 996,240 interactions, 22,916 students, and 2,750 questions.

4.3 Evaluation metrics

One way to measure how well the surrogate replicates the black box model is the R^2 measure as defined in [12]. It can be interpreted as the percentage of variance that is captured by the surrogate. When it is close to 1, we can say that the explainable model approximates well the behavior of the black box model. On the contrary, when it is close to 0 we can say that the explainable model fails to explain the black box model.

5 ON-GOING WORK

Some aspects need further research before a working implementation is ready. For instance, what is the best way to implement the mixup loss function or to choose values for the trade-off hyperparameter. Another aspect to explore is which of the candidate datasets can be used to test the approach.

Once the working implementation is ready, the next step to undertake will be the exploration of different trade-off curves to get insights about the strengths and weaknesses of the approach for different DKT and cognitive models.

ACKNOWLEDGMENTS

This work is framed in the context of the project HUMANAID (TED2021-129485B-C1) funded by MCIN/AEI/10.13039/501100011033 and the European Union "NextGenerationEU"/PRTR.

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